

Simulation of two spheres of Cesium nanoparticles extinction coefficient and its surface plasmon resonance (SPR)

Software used: DDSCAT 7.3.3

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By

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Contents

Contents	i
Abstract	1
Introduction	1
Methodology for setting up the necessary parameters for calculation in DDSCAT	2
Orientation of the target	3
Simulation Result	3
1 Simulation of Surface Plasmon Resonance (SPR) of two touching Cesium Nanospheres by varying the radius	3
2 Simulation of Surface Plasmon Resonance (SPR) of two Cesium Nanospheres by increasing separation	4
Conclusion	4
References	5

Abstract

Surface plasmon resonance (SPR) in cesium nanoparticles is a phenomenon where conduction electrons on the particle's surface resonate collectively with incident light at specific wavelengths. This resonance leads to enhanced light scattering and absorption, typically in the ultraviolet to visible range, depending on particle size, shape, and the surrounding medium. In this project, we are studying the plasmonic coupling of two spheres of cesium nanoparticles. We are using the Discrete Dipole Approximation[1] (DDA) method implemented in DDSCAT to simulate the variation in extinction coefficients at different radii and the distance of separation between two identical spheres. Unlike noble metals such as gold[3], we find for two cesium nanospheres, only one resonance peak is obtained around the wavelength of 700 *nm* to 710 *nm*.

Introduction

Cesium nanoparticles have emerged as a fascinating subject in nanophotonics and materials science due to their distinctive optical and electronic properties. Known for their strong plasmonic resonances, these nanoparticles exhibit unique interactions with light, particularly in the ultraviolet to visible spectrum. This makes them highly valuable for applications in sensing, imaging, and photonic devices. Cesium's high polarizability and its ability to sustain localized surface plasmon resonances (LSPR) enable exceptional control over light scattering and absorption processes, which are critical for advancing optical technologies. Furthermore, the tunable nature of cesium nanoparticles' optical responses based on size, shape, and surrounding medium adds to their versatility in designing tailored materials for cutting-edge applications.

The discrete dipole approximation (DDA) method, implemented through computational tools like DDSCAT, plays a vital role in studying the optical behavior of cesium nanoparticles. DDA provides a detailed simulation framework by representing the nanoparticle as an array of polarizable points, allowing accurate predictions of scattering, absorption, and extinction properties. This method is particularly advantageous for cesium, as it accommodates its complex dielectric behavior and the nanoscale dimensions of the particles. By enabling precise modeling of light interaction with irregularly shaped or non-homogeneous particles, DDA helps uncover insights into the mechanisms driving cesium's plasmonic phenomena. Such analyses are instrumental in optimizing their design for specific applications, ranging from biosensing to advanced nanophotonic systems.

Methodology for setting up the necessary parameters for calculation in DDSCAT

For two spheres, we use SPHERES_N method. We need to provide two important files:

1. Parameter file (in short par file): ddscat.par
2. Target file: filename.targ

In the par file, there are key places where we need to provide or change our inputs while most of the part remains of the file remains the same as in the example file of SPHERES_N in DDSCAT 7.3.3 Examples.

The following steps are used to set up the par file:

1. Setting up the SHPAR1 filename:

In SHPAR1, we need to provide the maximum extent in x-axis in terms dipole spacing d and target file. We need to set up the target file first (user guide[2]). In this file, the position of centre of the spheres are provided as x, y, z coordinates along with their radii in terms of an arbitrary unit D_0 . For our convenience, we set $a_0 = \frac{D_0}{2}$. So in the target file, the sphere radius is 0.5 in the units of D_0 . We then provide the maximum extent, N in x-axis in the units of D_0 and convert it in terms of d of the form $M*d$. The relation between them is

$$M * d = N * D_0 \quad (1)$$

For the known values of sphere radius, a_0 and dipole spacing, d , we use the following procedure:

- i The extent parameter N is calculated from the target file.
- ii As $D_0 = 2 * a_0$, we can rewrite (1) as

$$M = N * \frac{2 * a_0}{d} \quad (2)$$

We provide the value of M in the par file as:

M 1 'filename.targ' = SHPAR1 filename

2. Setting up the effective radii (in microns):

For n numbers of spheres of equal radii, we have effective radii, a_{eff} as:

$$a_{eff} = a_0 * n^{\frac{1}{3}} \quad (3)$$

Orientation of the target:

The incident wave is set to propagate in x-direction, i.e, $\mathbf{k}$ in $\hat{\mathbf{x}}$ with the polarization of the electric field, $\mathbf{E}$ in the $\hat{\mathbf{y}}$. The target frame is set such that the two spheres are oriented in $\hat{\mathbf{y}}$. The target frame is such that the coordinates are about the centers of the spheres. Writing d in terms of D_0 , we have the target orientation as:

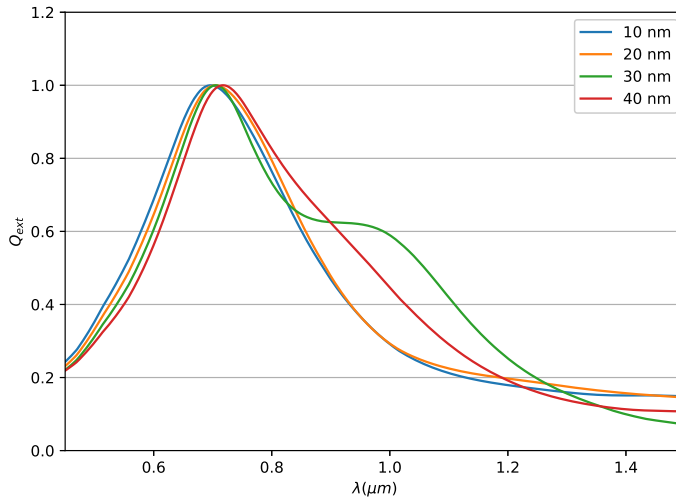
$$\begin{array}{cccc} \mathbf{x(j)} & \mathbf{y(j)} & \mathbf{z(j)} & \mathbf{a(j)} \\ 0 & 0 & 0 & 0.5 \\ 0 & 1+n*\frac{d}{D_0} & 0 & 0.5 \end{array}$$

Here, $n = 0,1,2,\dots$, $a(j)$ is the radius of j^{th} sphere and the extent in x-axis is from -0.5 to +0.5, hence $M = 1.0$. All the values of $x(j)$, $y(j)$, $z(j)$ and $a(j)$ considered here are in the units of D_0 .

Simulation Results:

1 Simulation of Surface Plasmon Resonance (SPR) of two touching Cesium Nanospheres by varying the radius

In this simulation we have focused on analysing the extinction coefficient, Q_{ext} and its dependencies with wavelength. The dipole spacing was set to $d=1.5 \text{ nm}$, ensuring a high-resolution representation of the particles. The radii of the nanoparticles were systematically varied at 10 nm , 20 nm , 30 nm and 40 nm to examine the size-dependent behavior of the extinction coefficient. The calculation was performed within the range of 300 nm to 1500 nm in the ambient medium as vacuum (with refractive index, $n=1.00$). The refractive index[4] was of cesium nanoparticles was taken from N. V. Smith. Optical constants[5]. It was observed that the radius of 30 nm has kind of two peaks.



Radius (nm)	Peak wavelength (nm)
10	697
20	705
30	705
40	719

Table 1: Radius and its corresponding wavelengths

Figure 1: Shifting of peak with the increasing radii

2 Simulation of Surface Plasmon Resonance (SPR) of two Cesium Nanospheres by increasing separation

In this simulation, we are varying the separation between the two spheres in the order of dipole spacing, $d = 1.5 \text{ nm}$. A fixed radius is used for all the separations, $a_0 = 10 \text{ nm}$. It was observed that only one peak is there for all the separations and they lie very closely around the wavelength 706 nm . With this result, we conclude that cesium has less plasmonic coupling even at very close spacing of dipoles. The lack of splitting in the extinction coefficient suggests that the intrinsic properties of cesium nanoparticles, such as their size and plasmonic behavior, may prevent the formation of coupled plasmon resonances at these scales.

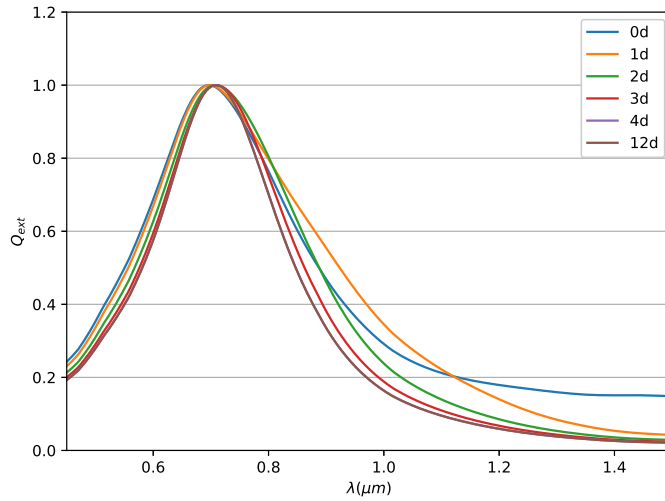


Figure 2: Extinction spectrum with the separation

Separation in terms of d ($d=1.5 \text{ nm}$)	Peak wavelength (nm)
0	697
1*d	698
2*d	707
3*d	709
4*d	707
12*d	707

Table 2: Radius and its corresponding wavelengths

Conclusion

The simulations revealed that varying the radius of cesium nanoparticles and the separation distance between them had minimal impact on the plasmonic response, as indicated by the absence of multiple peaks in the extinction spectrum. For all tested radii and separations, the extinction coefficient consistently showed a single peak in the range of 700 nm to 720 nm . This suggests that cesium's intrinsic optical properties and its plasmonic behavior are less sensitive to changes in particle size and inter-particle separation compared to other plasmonic materials like gold. The results highlight cesium's unique plasmonic characteristics and suggest limited coupling effects under the studied conditions.

References

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